

Software Requirements Specification

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Software Requirements Specification (SRS)

Project: E-Commerce Retail Platform

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System Overview

The E-Commerce Retail Platform is designed to facilitate the purchasing and selling of goods online. This platform will streamline retail operations by enabling users such as customers, suppliers, and administrators to interact with one another securely through an intuitive user interface.

Assumptions

1. The system will be accessible via a standard web browser on both Windows and macOS operating systems.
2. Users can log in using their unique email addresses or phone numbers.
3. All data entries (including product details, order information, etc.) are expected to be entered in the local time zone of the user's device.
4. Payment transactions will occur within a pre-determined credit/debit card validation period set by financial institutions.

Constraints

1. The system must comply with all relevant laws and regulations governing electronic commerce and data protection (e.g., GDPR, CCPA).
2. Support for legacy systems and devices is limited; only compatible hardware/software configurations are supported.
3. Data backup and recovery procedures will be initiated in the event of a major technical failure or security breach, but regular backups are not automated and must be manually triggered.
4. The system's scalability is restricted due to current infrastructure limitations and the need for performance optimization.

This SRS section serves as a foundation for establishing clear expectations and boundaries regarding both the capabilities and limits of the E-Commerce Retail Platform project.

Software Requirements Document

Introduction

This section outlines the requirements for developing the E-Commerce Retail Platform. The platform is designed to facilitate online retail transactions between buyers and sellers through an intuitive and user-friendly interface.

Scope

The scope of this project includes the development of a web-based application that acts as a marketplace connecting buyers and sellers, enabling them to transact products or services over the internet. It shall support all features defined in the Functional Requirements section for both Buyer and Seller actors.

Functional Requirements

Actors: ["Buyer", "Seller", "SupportAgent"]

Buyer Actor

- ****Browse Products:**** The platform allows a buyer to browse through various categories of products offered by sellers, view product details including images and descriptions, and select desired items.
- ****Add to Cart:**** A user can add selected products to their cart for easy checkout.
- ****Process Payment:**** After selecting the payment method, buyers are able to complete the purchase transaction securely. The platform supports multiple payment methods such as credit/debit cards, e-wallets, and other financial instruments compliant with industry standards.
- ****Resolve Dispute:**** In case of any disputes regarding products or services purchased, a buyer can initiate a dispute resolution process which involves communication with sellers for mutual understanding and negotiation.

Seller Actor

- ****Browse Products:**** Sellers are enabled to access their dashboard where they can manage products by adding new ones, editing existing listings, removing unwanted items from the inventory, etc. They also have visibility into sales data over time.
- ****Add to Cart:**** Suggested is an option for sellers to place orders directly through the platform's interface.
- ****Process Payment:**** Sellers are provided with a secure payment gateway to receive payments for their products or services ordered by buyers.
- ****Resolve Dispute:**** Sellers can initiate communication with customers regarding any disputes in relation to sales, offering resolutions and support to maintain customer satisfaction.

SupportAgent Actor

- ****Provide Support:**** The platform is equipped with an interface through which support agents can engage in dialogue with both buyer and seller for assistance during the transaction or while navigating various aspects of their accounts. This feature should facilitate a seamless user experience by providing timely, effective solutions to any issues encountered.
- ****Process Complaints/Feedback:**** A process for handling complaints and feedback from buyers and sellers is included within the system. Support agents are responsible for addressing these concerns in a fair and consistent manner.

This document outlines essential functionalities required for an E-Commerce Retail Platform. Each requirement is described with specific details to ensure comprehensive understanding of the platform's capabilities and limitations.

Non-Functional Requirements

Performance

1. ****Response Time:**** The system shall respond to user requests within 2 seconds under normal conditions.
2. ****Transaction Processing:**** Each transaction (such as adding an item to the cart or checking out) must be completed within 0.5 seconds.

Security

1. **Data Encryption**: Sensitive information such as credit card numbers and personal data will be encrypted using a strong encryption standard, AES-256.
2. **Authentication Mechanism**: Secure login mechanism should include both password hashing and two-factor authentication (e.g., SMS or email verification) for all accounts.
3. **Secure Communication**: All communication between the client and server shall use SSL/TLS protocols to ensure data integrity and confidentiality.

Usability

1. **User Experience**: The website shall be designed with user experience at its forefront, ensuring a pleasant browsing environment.
2. **Accessibility**: Ensure accessibility for users of all abilities through compliance with WCAG 2.0 guidelines.
3. **Responsive Design**: The interface must adapt to various devices and screen sizes (desktops, tablets, smartphones).

Reliability

1. **Fault Tolerance**: Implement redundancy in core services and systems that are mission-critical to avoid downtime.
2. **System Recovery Time**: In case of a system failure, it shall recover within 3 hours.

Scalability

1. **Horizontal Scaling**: Capable of handling increased user load by adding more servers or storage as needed.
2. **Vertical Scaling**: The capacity for the system to scale up through upgrades such as CPU, memory, and storage improvements should be possible.

Note: This document is provided in a clean Markdown format without any additional styling tags like HTML or CSS. Ensure that all non-functional requirements are clearly stated and fit within the constraints of the e-commerce domain and its standard complexity level.